

CONTACT
INFORMATION Chicago, IL, USA, 60617
E-mail: bpcoy@uchicago.edu

RESEARCH
INTERESTS

- **Planetary Habitability:** Earth-like exoplanet atmospheres and biosignatures
- **Exoplanet Spectroscopy:** Space-based transmission/emission spectroscopy of small exoplanets
- **Solar System Science:** Atmosphere and hydrological cycle systems of Titan

EDUCATION

University of Chicago, Chicago, IL

- PhD Student in Geophysical Sciences, Advisor: Prof. Edwin Kite, GPA: **4.0/4.0** 2023–
- M.S. in the [Physical Sciences Division](#) (GeoSci/Astro), GPA: **4.0/4.0** 2022
- Master’s Thesis: *The Role of Tectonic Luck and Weathering in Long-Term Climate Stability*

UCLA, Los Angeles, CA

2022–2023

- PhD Student in Planetary Science, Advisor: Prof. Jean-Luc Margot, GPA: **3.91/4.0**

UC Berkeley, Berkeley, CA

2015–2019

- B.A., [Astrophysics](#), GPA: **3.84/4.0**, *Cum Laude*, Dean’s List

EXTERNAL
RESEARCH
EXPERIENCE

- **NASA Jet Propulsion Laboratory**, Advisor: Dr. Keeyoon Sung Jul–Sept 2022
- **NASA Goddard Space Flight Center**, Advisor: Dr. Conor Nixon Jan–Jul 2021
- **NASA Marshall Space Flight Center**, Advisor: Danielle Moser Aug–Dec 2019
- **NASA Ames Research Center**, Advisor: Dr. Thomas Greene Jun–Aug 2019
- **UC Berkeley**, Advisor: Dr. Gaspard Duchêne Aug 2018–May 2019

FIRST-THIRD
AUTHOR
PUBLICATIONS

1. **Brandon Park Coy** et al. “Revealing the Shiny Nature of Lava Worlds: Small Planet Phase Curves with TESS”, 2026 (in review, *AJ*, arXiv:2609.26874)
2. **Brandon Park Coy**, Qiao Xue, Megan Weiner Mansfield et al., “Evidence for an Atmosphere on the Ultra-Short Period super-Earth HD 3167 b”, *ApJL*, 1005, L77, 2026
3. **Brandon Park Coy** et al. “Population-level Hypothesis Testing with Rocky Planet Emission Data: A Tentative Trend in the Brightness Temperatures of M-Earths”, *ApJ*, 987, 22, 2025.
4. **Brandon Park Coy**, Edwin Kite, and R.J. Graham. “The Role of Tectonic Luck in Long-term Habitability of Abiotic Earth-like Planets”, *PSJ*, 6(9), 218, 2025.
5. **Brandon Park Coy**, Conor Nixon, et al. “Spitzer IRS Observations of Titan as a Precursor to JWST MIRI Observations”. *PSJ*, Vol. 4 (6), p. 114, 2023.
6. Megan Weiner Mansfield, **Brandon Park Coy**, et al., “Evidence for the onset of atmospheres on lava planets”, 2026 (in prep, intended journal: *Nature*)
7. Marc Hon, Saul Rappaport, **Brandon Park Coy** et al., “Direct compositional sampling of fractionated material from an extant rocky exoplanet”, 2026 (in prep, intended journal: *Nature*)
8. Cole Smith, **Brandon Park Coy**, Megan Weiner Mansfield et al., “JWST MIRI Reveals a Potential Atmosphere on the Ultra-hot Rocky Planet TOI-431b”, accepted (*AJ*, arXiv:2609.03029), 2026
9. Sebastian Zieba, Laura Kreidberg, **Brandon Park Coy**, et al. “The dark and featureless surface of rocky exoplanet LHS 3844 b from JWST mid-infrared spectroscopy”, *Nat Astron*, 10, 1129, 2026.
10. Qiao Xue, Michael Zhang, **Brandon Park Coy**, et al. “The JWST Rocky Worlds DDT Program reveals GJ 3929b to likely be a bare rock”, *ApJL*, 995, L52, 2025.

11. Xuan Ji, Richard Chatterjee, **Brandon Park Coy** and Edwin Kite. “The Cosmic Shoreline Revisited: A Metric for Atmospheric Retention Informed by Hydrodynamic Escape”, *ApJ*, 992, 2, 2025.
12. Rafael Luque, **Brandon Park Coy**, Qiao Xue et al. “A dark, bare rock for TOI-1685 b from a JWST NIRSpec G395H phase curve”, *AJ*, 170, 49, 2025.
13. Conor Nixon, B Bézard*, T Cornet*, **Brandon Park Coy***, I de Pater*, M Es-Sayeh*, H Hammel*, E Lellouch*, N Lombardo*, M López-Puertas*, J Lora*, P Rannou*, S Rodriguez*, N Teanby*, E Turtle*, et al., “Titan’s Atmosphere in Late Northern Summer from JWST and Keck”. (*co-second author) *Nat Astron*, 9, 969, 2025.

CONTRIBUTED PUBLICATIONS

1. Courtney Dressing...**Brandon Park Coy** et al. “Early Exploration of the Scientific Discovery Space for the Habitable Worlds Observatory” (submitted, arXiv: 2608.11294)
2. Michael Zhang...**Brandon Park Coy** et al. “A carbon-rich atmosphere on a windy pulsar planet”, *The Astrophysical Journal Letters*, 995, L64, 2025
3. T. Bertin...Keeyoon Sung, **Brandon Park Coy** et al., “The HITRAN2024 methane update”, *JQSRT*, 349, 109736, 2025.
4. Megan Weiner Mansfield...**Brandon Park Coy** et al. “No Thick Atmosphere on the Terrestrial Exoplanet Gl 486b”, *ApJL*, 975, L22, 2024
5. Qiao Xue...**Brandon Park Coy** et al. “JWST Thermal Emission of the Terrestrial Exoplanet GJ 1132b”, *ApJL*, 973, L8, 2024
6. Jean-Luc Margot...**Brandon Park Coy** et al. “A Search for Technosignatures Around 11,680 Stars with the Green Bank Telescope at 1.15–1.73 GHz”. *AJ*, Vol. 166 (5), p. 206, 2023.
7. Gaspard Duchêne, J Oon, RJ De Rosa, P Kantorski, **Brandon Park Coy** et al., “A low-mass companion desert among intermediate-mass visual binaries: The scaled-up counterpart to the brown dwarf desert”. *MNRAS*, Vol. 519 (1), p. 778, 2023.
8. Edward Molter...Gaspard Duchêne, **Brandon Park Coy** et al. “Analysis of Neptune’s 2017 Bright Equatorial Storm”. *Icarus* Vol. 321 p. 324, 2019.

AWARDS/GRANTS

- **JWST Cycle 5 GO 10047 and 11694 Co-I**, NASA/STScI (\$20,000) 2026
- **JWST Cycle 4 GO 7407, 7875, and 8005 Co-I**, NASA/STScI 2025
- **Las Cumbres Observatory**, 2025A Observing Proposal PI, *Starspot Characterization of HD 207496* 2024
- **Illinois Space Grant Fellowship**, NASA/Illinois Space Grant Consortium (\$10,000) 2024
- **JWST Cycle 3 GO 6284 PI, 5263 Co-I**, NASA/STScI (\$150,000) 2024
- **McCormick Fellowship**, University of Chicago (\$4,500/year) 2023, 2024
- **Conference Travel Grant**, STScI (\$2,500) 2023
- **Incoming Student Fellowship**, UCLA (\$17,000) 2022

TALKS

- **Population-level Interpretation of Rocky Planet Surfaces and Atmospheres**
 - **Seminar** Max Planck Institute for Astronomy ExoCoffee Series 2026
 - **Seminar** Waterloo Centre for Astrophysics 2026
 - **Seminar** Carnegie EPL Astronomy Series 2026
 - **Seminar** STScI Exoplanets, Star, & Planet Formation Series 2026
 - **Seminar** University of Maryland Planetary Science Series 2026
- **Characterizing 10 Ultra-Hot Lava Worlds’ Thermal Emission with JWST**
 - **Talk** Other Worlds Laboratory Summer School (UCSC) 2026
 - **Talk** Exoplanets 6 (Porto, Portugal) 2026
 - **Talk** Rocky Worlds 4 Conference (Groningen, Netherlands) 2026
- **Atmospheric Theoretical Expectations for Rocky Worlds DDT Discoveries**

	◦ Invited Talk AAS 245 Rocky Worlds DDT Special Session 2025
	• Population-level Hypothesis Testing with Rocky Planet Emission Data: A Tentative Trend in the Brightness Temperatures of M-Earths ◦ Talk Atmospheric Escape and Replenishment Workshop (STScI) 2025 ◦ Talk Other Worlds Laboratory Summer School (UCSC) 2025 ◦ Talk Atmospheric Characterization of Rocky to Giant Exoplanets in Thermal Emission 2025 • What Sets the C/O Ratio in Sub-Neptune Atmospheres? ◦ Invited Talk Density Matters Workshop 2024 ◦ Lightning Talk Great Lakes Exoplanet Area Meeting 2023 • Spitzer IRS Observations of Titan as a Precursor to JWST MIRI Observations ◦ Talk NASA Outer Planets Assessment Group Meeting 2021, 2023 ◦ Talk NASA Astrobiology Institute Titan Meeting 2021 ◦ Talk International Symposium on Molecular Spectroscopy 2021
TEACHING AND MENTORING	• Undergraduate Research Advisor to Olivia Lucal 2026- ◦ <i>Constraining albedo, atmosphere dynamics, and tidal deformation of hot Jupiters with TESS</i> • English and Math Instructor Elite Educational Institute, Seoul, Korea Jan 2020–Dec 2020 • Undergraduate Student Instructor (Astro 9 & C10) UC Berkeley Jun–Dec 2018
SERVICE	• UChicago Exoplanet Journal Club Organizer 2025–2026 • Rocky Worlds 4 Breakout Session Chair (“Airless Planetary Surfaces: Moon, Mercury, and Beyond the Solar System”) 2026 • Volunteer, South Side Science Festival 2026- • Volunteer, Adler Planetarium 2011-2015, 2026- • Executive Officer, OASES (Oakland Asian Student Educational Services), UC Berkeley 2015-2019 • Reviewer for the Astrophysical Journal